

Assignment 7

Topic: CSS Layouts, Selectors, & Box Model

1. Borders and Padding Demonstration

INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head><link rel="stylesheet" href="style.css"></head>
<body>
  <header><h1>Border & Padding Demo</h1></header>
  <main class="box-container">
    <section class="box solid-box">Solid Border</section>
    <section class="box dashed-box">Dashed Border</section>
    <section class="box double-box">Double Border</section>
    <section class="box groove-box">Groove Border</section>
  </main>
  <footer><p>Footer Content</p></footer>
</body>
</html>
```

STYLE.CSS

```
body { font-family: sans-serif; margin: 0; }
header, footer { background: #333; color: white; text-align: center; padding: 15px; }
.box-container {
  display: flex;
  justify-content: center;
  gap: 20px;
  padding: 50px;
}
.box {
  /* Padding values to demonstrate internal spacing */
  padding: 30px 20px;
  background: #f4f4f4;
  border-radius: 15px; /* Rounded corners */
  border-width: 4px;
  transition: border-color 0.3s ease;
}
.solid-box { border-style: solid; border-color: #e74c3c; }
.dashed-box { border-style: dashed; border-color: #3498db; }
.double-box { border-style: double; border-color: #2ecc71; }
.groove-box { border-style: groove; border-color: #9b59b6; }

.box:hover { border-color: #f1c40f; } /* Hover effect */
```

2. Height and Width Practice Layout

INDEX.HTML

```
<main class="practice-layout">
  <div class="box box-1">Box 1 (px & vh)</div>
  <div class="box box-2">Box 2 (%)</div>
  <div class="box box-3">Box 3 (em) <br> Overflow testing content goes here
to show scroll.</div>
</main>
```

STYLE.CSS

```
.practice-layout {
  display: flex;
  justify-content: center;
  align-items: center; /* Centered vertically and horizontally */
  min-height: 100vh;
  gap: 20px;
  background: #ecf0f1;
}
.box {
  padding: 20px;
  color: white;
  font-family: sans-serif;
  text-align: center;
}
.box-1 {
  width: 200px;
  height: 30vh;
  background: #e74c3c;
}
.box-2 {
  width: 25%; /* Responsive width */
  max-width: 300px; /* Limits max width */
  min-height: 150px; /* Ensures a minimum height */
  background: #3498db;
}
.box-3 {
  width: 15em;
  height: 150px;
  background: #2ecc71;
  overflow: auto; /* Adds scroll if content overflows */
}
```

3. Gradient Background Webpage

INDEX.HTML

```
<body>
  <header class="fixed-header">Gradient Showcase</header>
  <main>
    <section class="linear-section">
      <h2>Linear Gradient Background</h2>
      <p>Parallax scrolling effect active.</p>
    </section>
    <section class="radial-section">
      <h2>Radial Gradient Background</h2>
    </section>
  </main>
  <footer class="fixed-footer">Footer</footer>
</body>
```

STYLE.CSS

```
body { margin: 0; padding-top: 60px; padding-bottom: 60px; font-family: sans-serif; }
.fixed-header {
  position: fixed; top: 0; width: 100%; height: 60px;
  background: #222; color: white; display: flex; align-items: center;
  justify-content: center; z-index: 100;
}
.fixed-footer {
  position: fixed; bottom: 0; width: 100%; height: 60px;
  background: #222; color: white; display: flex; align-items: center;
  justify-content: center;
}
section {
  height: 100vh;
  display: flex; flex-direction: column; justify-content: center; align-items: center;
  color: white; font-size: 1.5rem; text-shadow: 1px 1px 3px rgba(0,0,0,0.5);
}
.linear-section {
  /* Linear gradient with fixed attachment for parallax */
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  background-attachment: fixed;
}
.radial-section {
  /* Radial gradient */
}
```

```
background: radial-gradient(circle, #ff9a9e 0%, #fecfef 99%, #fecfef 100%);  
color: #333;  
text-shadow: none;  
}
```

4. Shadows and Depth Effect Layout

INDEX.HTML

```
<main class="card-container">
  <article class="profile-card card-1">
    <h2>Card 1</h2>
    <p>Subtle shadow</p>
  </article>
  <article class="profile-card card-2">
    <h2>Card 2</h2>
    <p>Medium shadow</p>
  </article>
  <article class="profile-card card-3">
    <h2>Card 3</h2>
    <p>Harsh shadow</p>
  </article>
</main>
```

STYLE.CSS

```
.card-container {
  display: flex; /* Single row arrangement */
  justify-content: center;
  gap: 30px;
  padding: 50px;
  background: #fdfdfd;
}
.profile-card {
  background: white;
  padding: 40px 30px;
  border-radius: 10px;
  text-align: center;
  font-family: sans-serif;
  transition: transform 0.3s ease, box-shadow 0.3s ease; /* Smooth animation */
}
.profile-card h2 {
  text-shadow: 2px 2px 4px rgba(0,0,0,0.2); /* Text shadow for depth */
}
/* Different shadow styles */
.card-1 { box-shadow: 0 4px 6px rgba(0,0,0,0.1); }
.card-2 { box-shadow: 0 10px 15px rgba(0,0,0,0.15); }
.card-3 { box-shadow: 5px 5px 0px rgba(0,0,0,0.8); }
```

```
/* Hover effect lifting up */  
.profile-card:hover {  
  transform: translateY(-10px);  
  box-shadow: 0 20px 25px rgba(0,0,0,0.2);  
}  
.card-3:hover { box-shadow: 10px 10px 0px rgba(0,0,0,0.8); }
```

5. Advanced CSS Selectors

INDEX.HTML

```
<section class="selectors-demo">
  <h2>Advanced Selectors</h2>
  <p>This is a sibling paragraph.</p>
  <ul class="item-list">
    <li><a href="https://example.com">Secure Link</a></li>
    <li><a href="http://example.com">Insecure Link</a></li>
    <li><input type="text" placeholder="Text Input"></li>
    <li><input type="password" placeholder="Password Input"></li>
  </ul>
</section>
```

STYLE.CSS

```
.selectors-demo { font-family: sans-serif; padding: 20px; }

/* 1. Combinators */
/* Adjacent sibling (+) */
h2 + p { color: #555; font-style: italic; }
/* Child combinator (>) */
.item-list > li { margin-bottom: 10px; list-style-type: none; }

/* 2. Attribute Selectors */
/* Selects inputs explicitly with type="text" */
input[type="text"] { border: 2px solid #3498db; padding: 5px; }
/* Selects anchor tags whose href starts with https */
a[href^="https"] { color: #2ecc71; font-weight: bold; }

/* 3. Pseudo-classes */
/* :hover state */
a:hover { text-decoration: none; color: #e74c3c; }
/* :focus state */
input:focus { outline: 2px dashed #9b59b6; }
/* :nth-child for alternating styles */
.item-list li:nth-child(even) { background: #f9f9f9; padding: 5px; }

/* 4. Pseudo-elements */
/* ::before adds content before element */
a[href^="https"]::before { content: "🔒 "; }
/* ::after adds content after element */
```

```
.item-list::after { content: "End of list."; display: block; margin-top: 10px;
font-size: 0.8em; color: #aaa; }
```

6. CSS Transformations Showcase

INDEX.HTML

```
<main class="transform-container">
  <div class="transform-box">
    <div class="box t-rotate">Rotate</div>
    <p>rotate(45deg)</p>
  </div>
  <div class="transform-box">
    <div class="box t-scale">Scale</div>
    <p>scale(1.5)</p>
  </div>
  <div class="transform-box">
    <div class="box t-translate">Translate</div>
    <p>translate(20px, -20px)</p>
  </div>
  <div class="transform-box">
    <div class="box t-skew">Skew</div>
    <p>skew(20deg)</p>
  </div>
</main>
```

STYLE.CSS

```
.transform-container {
  display: flex; /* Arrange horizontally */
  justify-content: center;
  gap: 40px;
  padding: 60px;
  font-family: sans-serif;
  text-align: center;
}
.box {
  width: 100px;
  height: 100px;
  background: #3498db;
  color: white;
  display: flex;
  align-items: center;
  justify-content: center;
  margin-bottom: 15px;
  transition: all 0.4s ease; /* Required transition */
}
```

```
/* Transformations applied on hover */  
.t-rotate:hover { transform: rotate(45deg); background: #e74c3c; }  
.t-scale:hover { transform: scale(1.5); background: #2ecc71; }  
.t-translate:hover { transform: translate(20px, -20px); background: #9b59b6; }  
.t-skew:hover { transform: skew(20deg); background: #f1c40f; color: #333; }
```

7. Typography and Text Styling Page

INDEX.HTML

```
<article class="typography-demo">
  <h2>The Art of Typography</h2>
  <p>Typography is the <strong>foundation</strong> of readable web design. By
  carefully adjusting fonts, we ensure that users can <mark>consume content</
  mark> without eye strain.</p>
  <p>Using <em>appropriate whitespace</em> creates visual balance.</p>
</article>
```

STYLE.CSS

```
.typography-demo {
  /* Different font families and sizes */
  font-family: 'Georgia', serif;
  font-size: 18px;
  /* Line height and letter spacing for readability */
  line-height: 1.8;
  letter-spacing: 0.5px;
  text-align: justify;
  max-width: 600px;
  margin: 40px auto;
}

.typography-demo h2 {
  font-family: 'Helvetica Neue', sans-serif;
  font-weight: 700;
  font-size: 2.5rem;
  text-align: center;
}

/* Custom styling for inline elements */
mark { background-color: #ffeaa7; color: #d63031; padding: 0 4px; }
strong { font-weight: 900; color: #0984e3; }
em { font-style: italic; color: #00b894; }

/* Pseudo-elements for emphasis */
.typography-demo p:first-of-type::first-letter {
  font-size: 3rem;
  float: left;
  margin-right: 8px;
  line-height: 1;
  color: #2d3436;
  font-weight: bold;
```

```
}  
.typography-demo p:first-of-type::first-line {  
  font-variant: small-caps;  
  color: #636e72;  
}
```

8. Box Model Exploration Page

INDEX.HTML

```
<main class="box-model-demo">
  <div class="margin-layer">
    <span>Margin</span>
    <div class="border-layer">
      <span>Border</span>
      <div class="padding-layer">
        <span>Padding</span>
        <div class="content-layer">
          <span>Content (border-box applied)</span>
        </div>
      </div>
    </div>
  </div>
</main>
```

STYLE.CSS

```
.box-model-demo {
  display: flex;
  justify-content: center;
  padding: 50px;
  font-family: monospace;
  text-align: center;
}
.margin-layer {
  background: #f9ca24; /* Yellow margin */
  padding: 30px;
  outline: 2px dashed #f0932b;
}
.border-layer {
  background: #eb4d4b; /* Red border */
  padding: 10px; /* Represents border width physically here */
  outline: 2px solid #ff7979;
}
.padding-layer {
  background: #badc58; /* Green padding */
  padding: 40px;
  outline: 2px dotted #6ab04c;
}
.content-layer {
  background: #7ed6df; /* Blue content */
}
```

```
width: 200px;
height: 100px;
display: flex;
align-items: center;
justify-content: center;
/* Demonstrating box-sizing */
box-sizing: border-box;
border: 5px solid #22a6b3;
}
span { font-weight: bold; color: #130f40; }
```

9. Outline and Focus Styling Form

INDEX.HTML

```
<form class="styled-form">
  <div class="form-group">
    <label>Name:</label>
    <input type="text" class="focus-input" placeholder="Enter name">
  </div>
  <div class="form-group">
    <label>Message:</label>
    <textarea class="focus-input" rows="4" placeholder="Your message"></
textarea>
  </div>
  <button type="button">Submit</button>
</form>
```

STYLE.CSS

```
.styled-form {
  max-width: 400px;
  margin: 50px auto;
  font-family: sans-serif;
}

.form-group {
  margin-bottom: 20px;
  display: flex;
  flex-direction: column;
}

label { margin-bottom: 8px; font-weight: bold; color: #333; }

.focus-input {
  padding: 12px;
  border: 1px solid #ccc;
  border-radius: 6px;
  box-shadow: inset 0 2px 4px rgba(0,0,0,0.05); /* Subtle inner shadow */

  /* Outline settings */
  outline-style: solid;
  outline-width: 2px;
  outline-color: transparent;
  outline-offset: 4px;
  transition: outline-color 0.3s, border-color 0.3s;
}
```

```
/* Outline changes on focus */
.focus-input:focus {
  outline-color: #3498db;
  border-color: #3498db;
}

button {
  padding: 12px 20px;
  background: #2c3e50;
  color: white;
  border: none;
  border-radius: 6px;
  cursor: pointer;
}
```

10. Navigation Bar with Transitions and Hover Effects

INDEX.HTML

```
<body>
  <nav class="navbar">
    <a href="#" class="nav-link active">Home</a>
    <a href="#" class="nav-link">About</a>
    <a href="#" class="nav-link">Services</a>
    <a href="#" class="nav-link">Portfolio</a>
    <a href="#" class="nav-link">Contact</a>
  </nav>
  <main style="min-height: 80vh; padding: 20px;">Content goes here.</main>
  <footer class="simple-footer">
    <p>Simple Centered Text Footer</p>
  </footer>
</body>
```

STYLE.CSS

```
body { margin: 0; font-family: sans-serif; }
.navbar {
  display: flex;
  justify-content: center;
  background: #34495e;
  padding: 10px;
}
.nav-link {
  color: white;
  text-decoration: none;
  padding: 10px 20px;
  margin: 0 5px;
  border: 2px solid transparent;
  border-radius: 4px;
  transition: all 0.3s ease; /* Smooth transitions */
}
/* Hover effect */
.nav-link:hover {
  border-color: #1abc9c;
  background: rgba(26, 188, 156, 0.1);
}
/* Highlight active link */
.nav-link.active {
  background: #1abc9c;
  color: #fff;
}
```

```
    border-color: #1abc9c;
}

.simple-footer {
    text-align: center;
    background: #2c3e50;
    color: white;
    padding: 20px;
}

/* Responsive vertical stack */
@media (max-width: 700px) {
    .navbar {
        flex-direction: column;
        align-items: center;
    }
    .nav-link {
        width: 100%;
        text-align: center;
        margin: 5px 0;
        box-sizing: border-box;
    }
}
```